KAZI ABRAR MAHMUD

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Research Portfolio | Google Scholar

Education

Bangladesh University of Engineering and Technology *February 2020 - February 2025*
BSc. in Electrical and Electronic Engineering Dhaka, Bangladesh

- **Major**: Communication and Signal Processing (CSP)
- **CGPA**: 3.94/4.00
- Achieved 4.00/4.00 in the final semester of my undergraduate studies

Adamjee Cantonment College *June 2019*
HSC in Science Dhaka, Bangladesh

- **GPA**: 5.00/5.00 | **Award**: Govt. Merit Scholarship for outstanding HSC performance

Coursework

Self-Study Courses

- Data Structure & Algorithm (certificate)
- Deep Learning (personal notes & projects)
- Broadband Communication (personal notes)
- Medical Imaging (personal notes)
- Biomedical Signal Processing (personal notes)

Relevant Coursework

- Random Signals and Processes
- Robotics and Automation
- Digital Image Processing
- Microprocessor and Embedded Systems
- Radar and Satellite Communication

Research Interests

- Audio Signal Processing
- Computer Vision
- LLM based Robotics
- Large Language Model
- Deep learning
- Autonomous Navigation

Publications

Full list on Google Scholar. Author name in **bold** denotes myself.

Journal Articles

[J1] Shadman Sobhan, **Kazi Abrar Mahmud**, Abduz Zami, Mohiuddin Ahmed, and Md Palash Uddin. "MedPrompt: LLM–CNN Fusion with Weight Routing for Medical Image Segmentation and Classification." *Biomedical Signal Processing and Control*, vol. 113, article 109251, 2026. **(Q1 Journal)** [DOI]

Conference Proceedings

[C1] **Kazi Abrar Mahmud**, Fahim Ahmed, Rayhan Siddiq, and Shadman Sobhan. "Edge-Optimized Firefighting Robot: Autonomous Navigation via PID Control and Fire Detection with FOMO Vision." In *Proc. 2025 7th International Conference on Electrical Information and Communication Technology (EICT)*, 2025. IEEE Xplore. [DOI]

Under Review / Preprints

[P1] **Kazi Abrar Mahmud**, N. K. Dhurubo, T. Kirttonia, S. H. Ujjal, and M. A. Haque. "Bridging Thought and Action: Taming Long-Horizon Instability in Open-Source LLM Agents with a MetaTool-Enhanced ROS Framework." *arXiv preprint arXiv:2609.13335*, 2026. **(Under review at a Q1 journal)** [arXiv]

Research Experience

- **Undergraduate Thesis: Towards Empathetic Voice Assistants – Enhancing Long-Term Conversations with Small Language Models, Semantic Routing, and Emotion-Aware Speech Recognition.**

  Supervisor: Dr. Mohammad Ariful Haque *Completion Date: March 2025*

  - Fine-tuned a Small Language Model (SLM) and integrated semantic routing to sustain long-term conversations while reducing inference time.
  - Developed a custom Emotion-Aware Speech Recognition model to enrich the conversational context available to the SLM.

- **Research Project: Equilibrium Matching-based Policy for Robot Behavioral Cloning.**
  Supervisor: Dr. Mohammad Ariful Haque *In Progress*
  - Developing a novel behavioral cloning policy built on Equilibrium Matching that learns an implicit energy landscape over actions, as an alternative to diffusion-based policies.
  - Achieved faster training convergence than iterative denoising-based policy learning.
  - Demonstrated strong task performance when running the policy in implicit energy mode.

Research Projects

- **3-DOF Arm Imitation Learning via Diffusion Policy in Gazebo.**
  - Built an end-to-end imitation learning pipeline on a custom 3-DOF linear arm (URDF) in Gazebo/ROS: recorded expert demonstrations for behavioral cloning, converted raw ROS data into the Hugging Face LeRobot dataset format, and trained a Diffusion Policy. Deployed the policy through an inference server that drives the simulated arm autonomously.

- **AgroBot: An Intelligent Vision-Guided System for Sustainable Weed Detection & Elimination.**
  Supervisor: Dr. Celia Shahnaz
  - Developed an intelligent vision-guided system using ROS on a Raspberry Pi 5, integrating a fine-tuned vision model with an autonomous navigation system to traverse crop fields and eliminate weeds efficiently.

- **Analytical Solution Generation for n-DOF Manipulators with a Processing 3 Joint-State Visualizer.**
  - Developed a tool that composes homogeneous transformation matrices from input DH parameters, enabling analytical solutions for higher-order robotic arm manipulators, and simulated the resulting joint-state output in a Processing 3 visualizer.

- **Numerical Analyzer for Poisson's Equation of Gravitational Potential in 2D Space.**
  - Implemented a numerical solver for Poisson's equation of gravitational potential, with an interactive interface that visualizes the gravitational fields of multi-body systems.

- **Direct-Conversion Receiver (DCR) based AM Radio Station.**
  - Implemented DSB signal transmission and Direct Conversion Receiver reception, optimized the design through Eagle schematic simulation, and built the hardware to establish wireless communication.

Teaching Experience

- **United International University (UIU) [🌐]** *July 2025 - present*
  Lecturer United City, Madani Ave
  - Deliver theoretical and hands-on lectures to undergraduate students, emphasizing conceptual understanding and practical implementation.
  - Mentored a capstone (final-year project) group that designed and built a morphing drone, modeled and analyzed its dynamics, and implemented a low-level controller-based flight control system.
  - Serve as Moderator of the Electrical and Electronic Club (UIU EEC), overseeing activities, projects, and student engagement.
  - Coordinated the launch of Power Energy Training Academy (PETA) training programs focused on Industry 4.0 technologies.

Honors and Awards

- **1st Runner-Up, Poster Presentation, BEAR Summit 2025:** "AgroBot: AI-Powered Precision Weeding" (Supervisor: Dr. Celia Shahnaz) 2025
- **University Merit Scholarship:** Academic honour by BUET 2020–2025
- **Dean's List Award:** Top 10% at every level, every semester, BUET 2021–2025
- **Undergraduate CSP Major Merit Position:** 1st among 65 candidates of the Communication and Signal Processing Major
- **Govt. Merit Scholarship:** Ministry of Education, Bangladesh (HSC, Dhaka Board) 2019

ACADEMIC SERVICE & MENTORSHIP

- **Service: Official Reviewer, BEAR Summit 2025 [📄]** *16 July 2025*
  Robotics Session, invited by the Bangladesh Computer Council National Science & Technology Complex
  - Reviewed the robotics session, "*Robotics for Resilience, Labor, and Industry Revolution 4.0 and 5.0*", at the national summit on Biotech, Electronics, AI, and Robotics (BEAR), by invitation of the executing agency for the ICT Division.
- **Mentorship: Workshop Instructor, Robotic Boot Camp [🌐]** *June 2025*
  Under Dr. Mohammad Ariful Haque IRAB, BUET, Dhaka
  - Taught robot kinematics lectures and labs, and led the final project presentation.
- **Mentorship: Lead Instructor, RoboHack Workshop [🌐]**
  IEEE Robotics & Automation Society BUET, Dhaka, Bangladesh
  - Led hands-on sessions on building smart mobile robots using ROS, Raspberry Pi, and ESP32.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Verilog, MATLAB, Processing 3, ROS
- **Hardware Skill:** STM32 boards, ESP & Arduino boards, FPGA
- **Circuit Simulation and Design:** PSpice, Keil uVision, Autodesk Eagle PCB
- **Network Simulation and Design:** Cisco Packet Tracer
- **Others:** Linux, Autodesk Fusion 360, AutoCAD, Overleaf (LaTeX), DaVinci Resolve

REFERENCES

Dr. Mohammad Ariful Haque
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Dr. Farseem Mannan Mohammedy
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Relationship: ***Academic Advisor***